

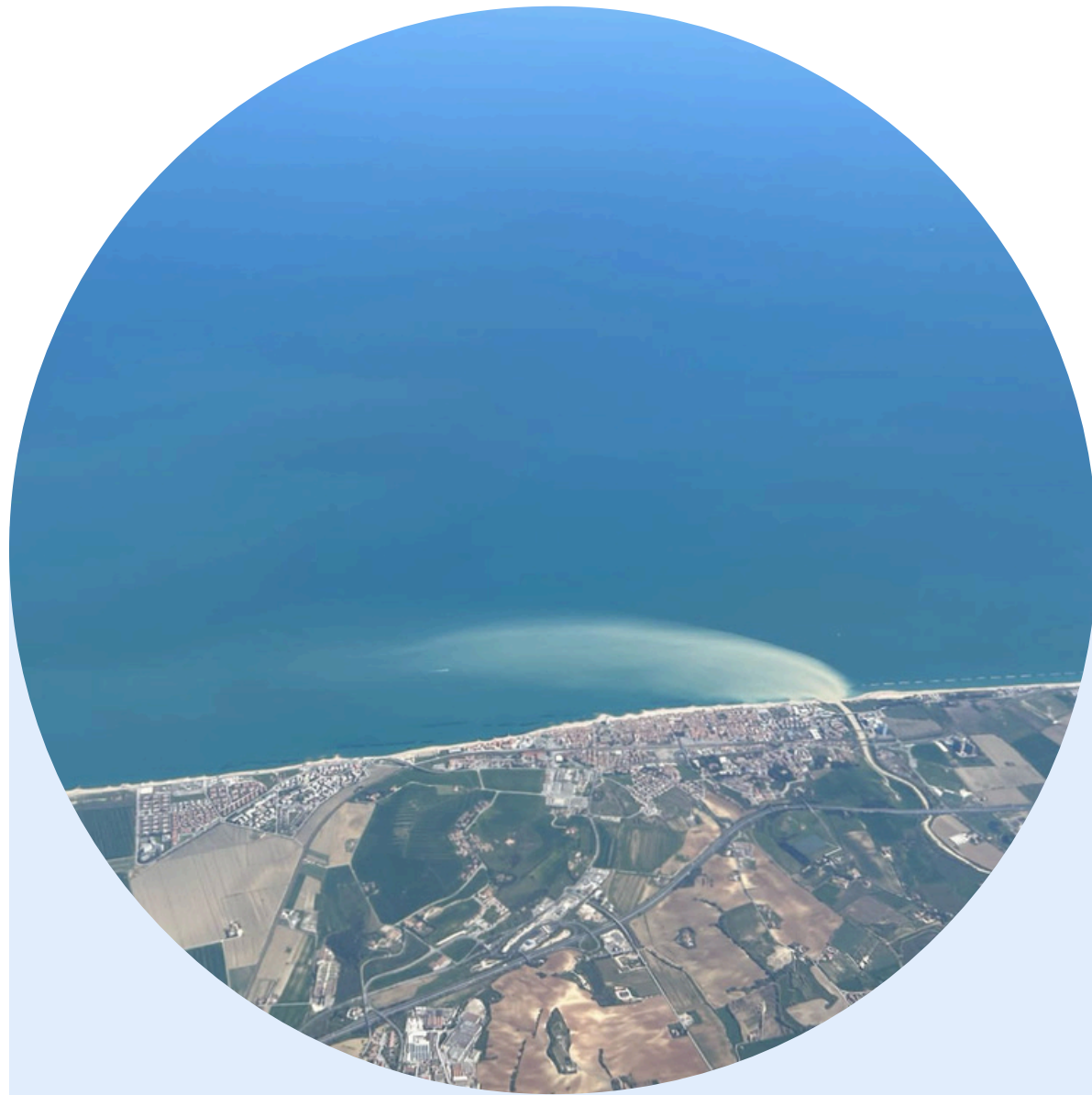
Field practices: Marine Monitoring

CASE STUDY: WATER CURRENT IMPACT ON THE CIVITANOVA COAST

**Calocero Andrea, Savino Eleonora C., Sirianni Giulia,
Sturiale Letizia B., Tessari Alessandro**



Introduction



The relationship between humans and rivers has been in the past of fundamental importance for the progress of human civilization. However, continual anthropogenic pressure have led to a progressive deterioration of the water quality with rivers that can be sources of both natural and anthropogenic pollution.

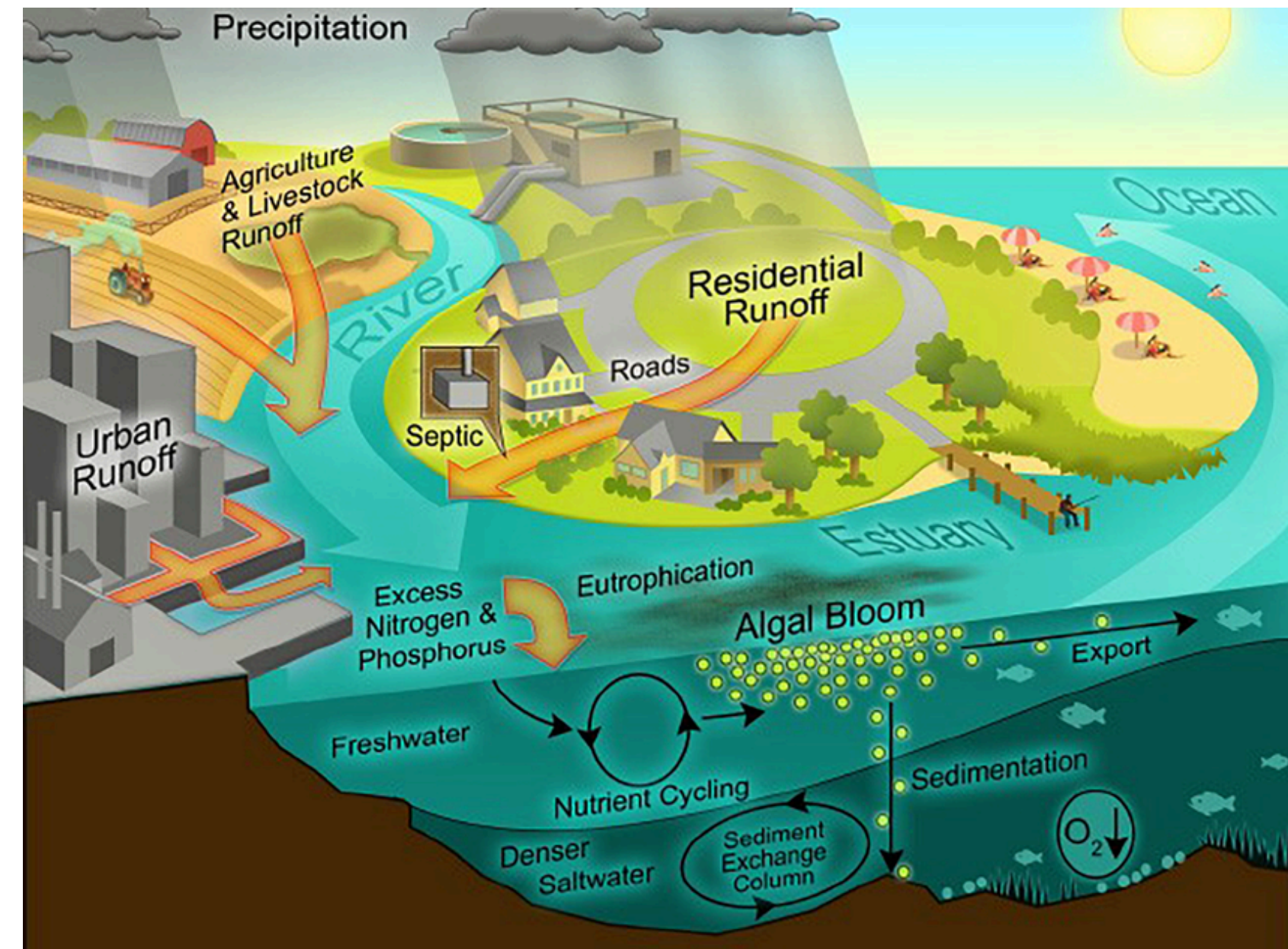
The presence of river runoff can influence the achievement of the Good Environmental Status of sea water, according to the MSFD 56/2008



The Marine Strategy Framework Directive

Rivers impacts on marine ecosystems

- **INORGANIC NUTRIENTS INPUTS** coming from agriculture and urban runoff due to fertilizers, detergents, etc. (D5)
- **ORGANIC MATTER INPUTS** coming from the land system, called secondary organic matter (not of autotrophic origin) (D5)
- In case of **EUTROPHICATION** the area can be affected by **BIODIVERSITY LOSS**, mainly due to deoxygenation with alteration of the of the communities structure with replacement of sensitive species with more resistant, often opportunistic ones (D5 - D1)



Malone, Thomas & Newton, Alice. (2020). The Globalization of Cultural Eutrophication in the Coastal Ocean: Causes and Consequences. *Frontiers in Marine Science*. 7. 10.3389/fmars.2020.00670.



CONTAMINANTS INPUTS can be carried out by rivers, releasing into the marine ecosystem elements as plastics, microplastics and other contaminants, which can be directly discharged on them or washed by rain from the terrestrial system, as hydrocarbons and heavy metals (**D8 - D10**)

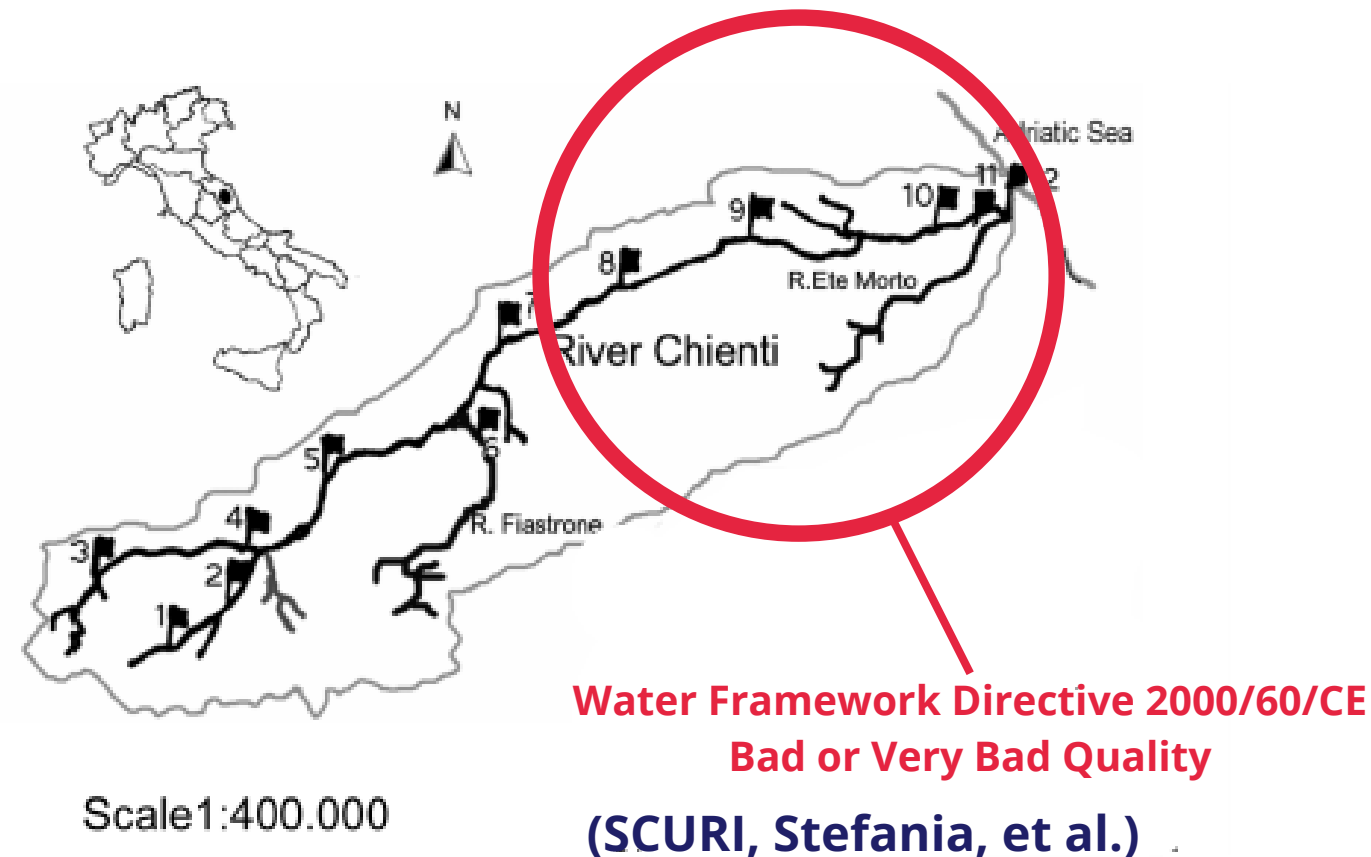


This contaminants can be **BIOACCUMULATED** in marine organisms and **BIO-MAGNIFICATED** along the trophic web with the possibility to reach humans too (**D9**)



Chienti Rivers

Chienti River is a watercourse of Central Appennines, it is 91 km long with maximum flow rate occurs in spring and minimum in late summer.



The Western Adriatic Coastal Current (**WACC**) flows mainly **southward**. It transports sediments and nutrients from riverine sources. While **Bora** winds **strengthen** this southward flow, **Scirocco** winds can temporarily **slow it down**.



Courtney K. Harris,¹ Christopher R. Sherwood, Richard P. Signell, Aaron J. Bever, and John C. Warner (2008). Sediment dispersal in the northwestern Adriatic Sea. JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 113, C11503, doi: 10.1029/2006JC003868

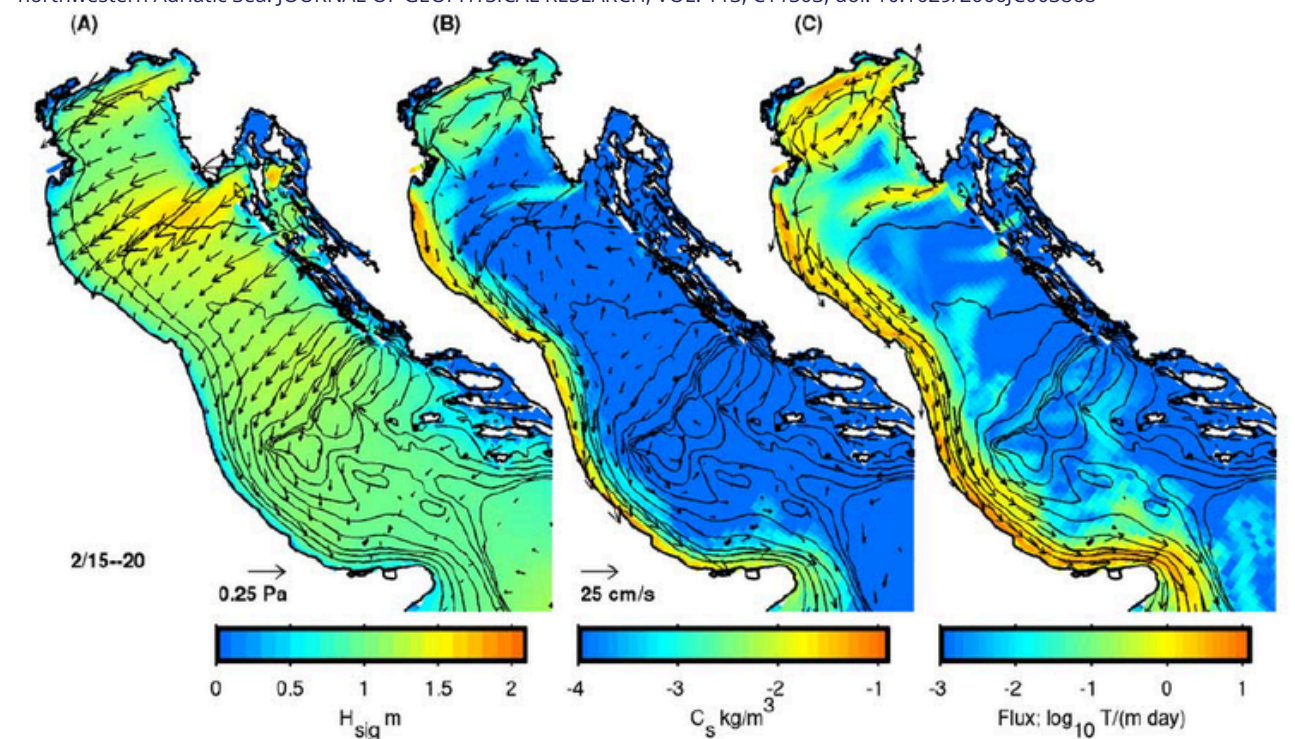


Figure 13. Conditions averaged during a Bora from 15 to 20 February 2003. (a) Wind stress (arrows; see scale for magnitude) and wave height (color). (b) Depth- and time-averaged sediment concentration (color) and current velocity (arrows; see scale for magnitude). (c) Time-averaged, depth-integrated flux ($\text{t m}^{-2} \text{d}^{-1}$). Arrows show direction where flux exceeds $0.1 \text{ t m}^{-2} \text{d}^{-1}$. Contours drawn every 25 m up to 200-m water depth.

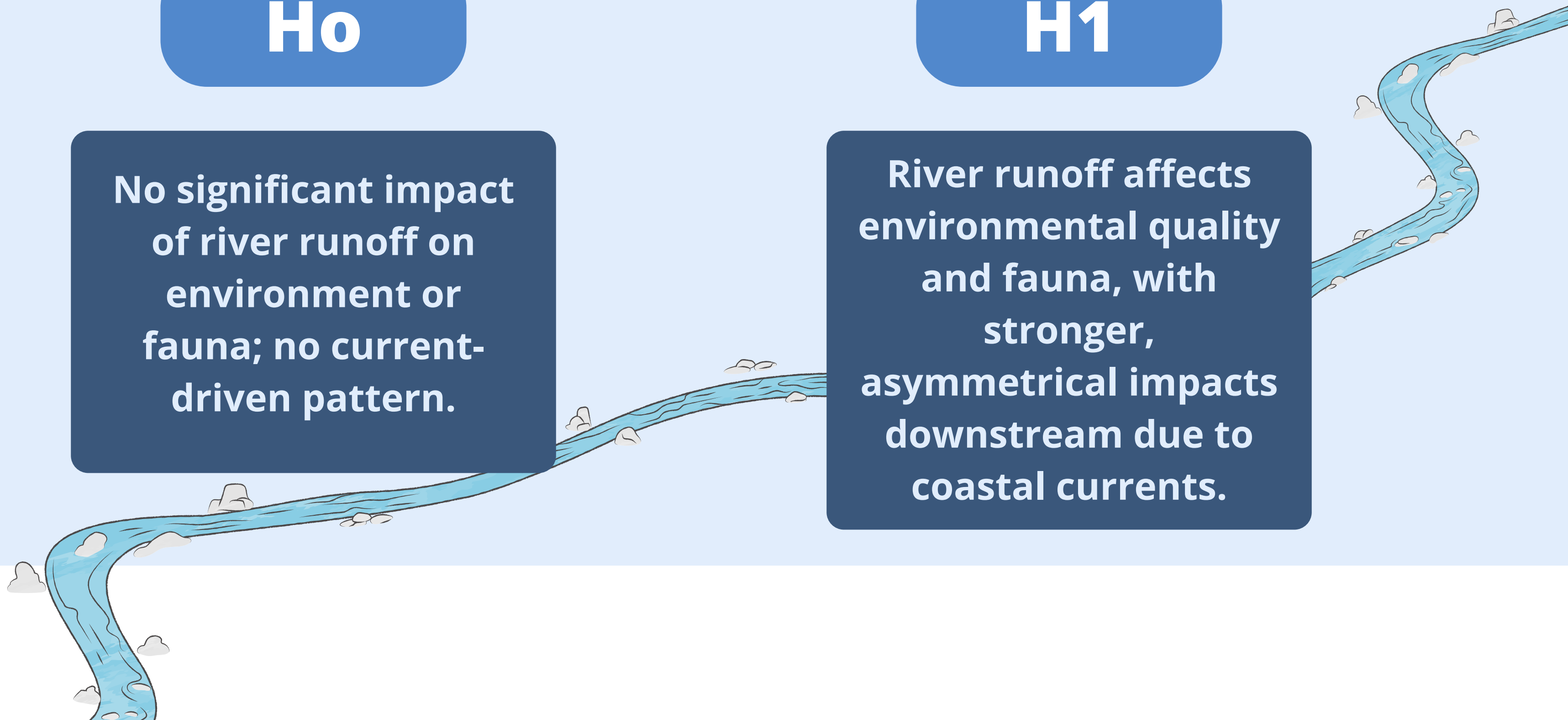
Hypotheses

H₀

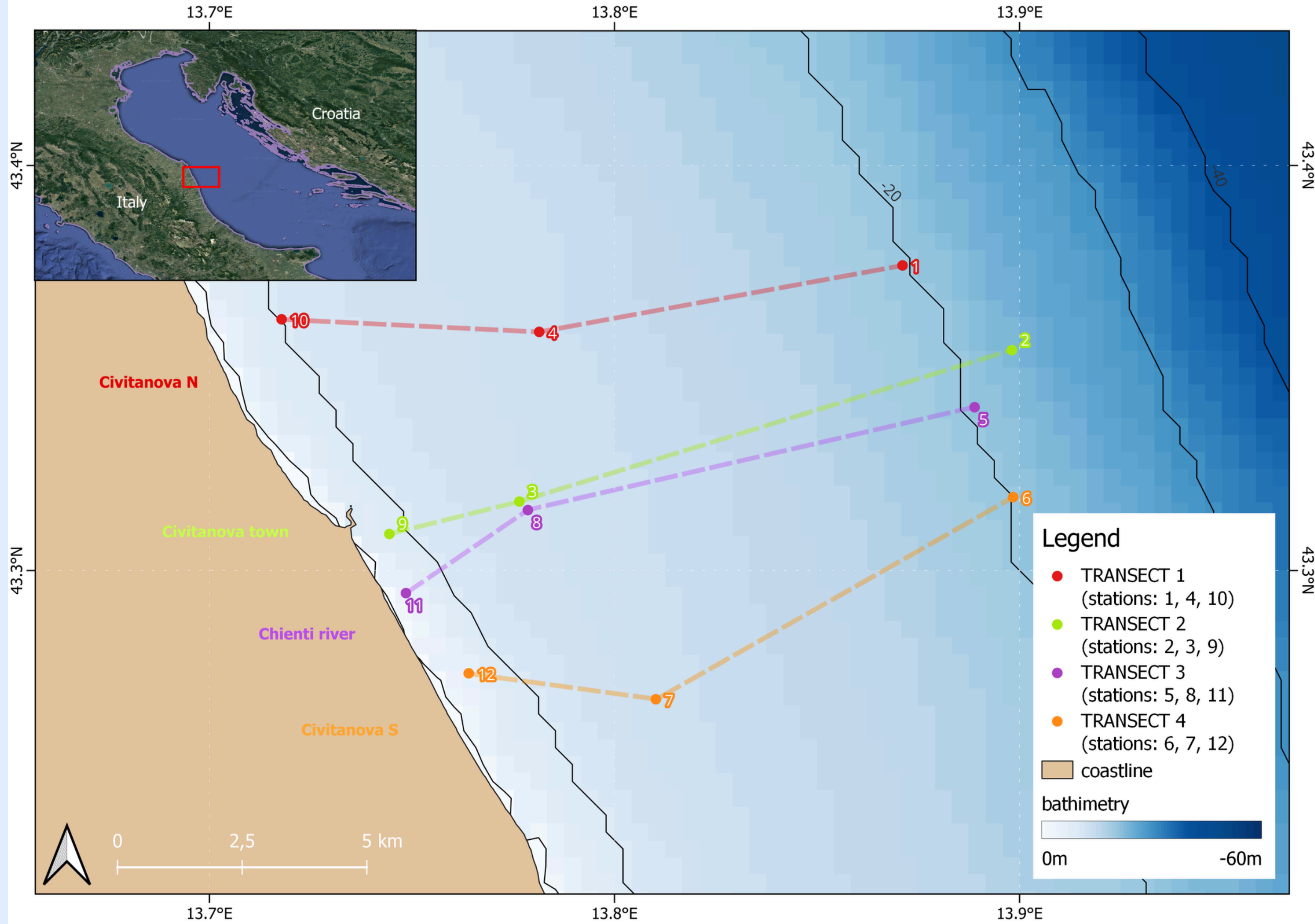
No significant impact of river runoff on environment or fauna; no current-driven pattern.

H₁

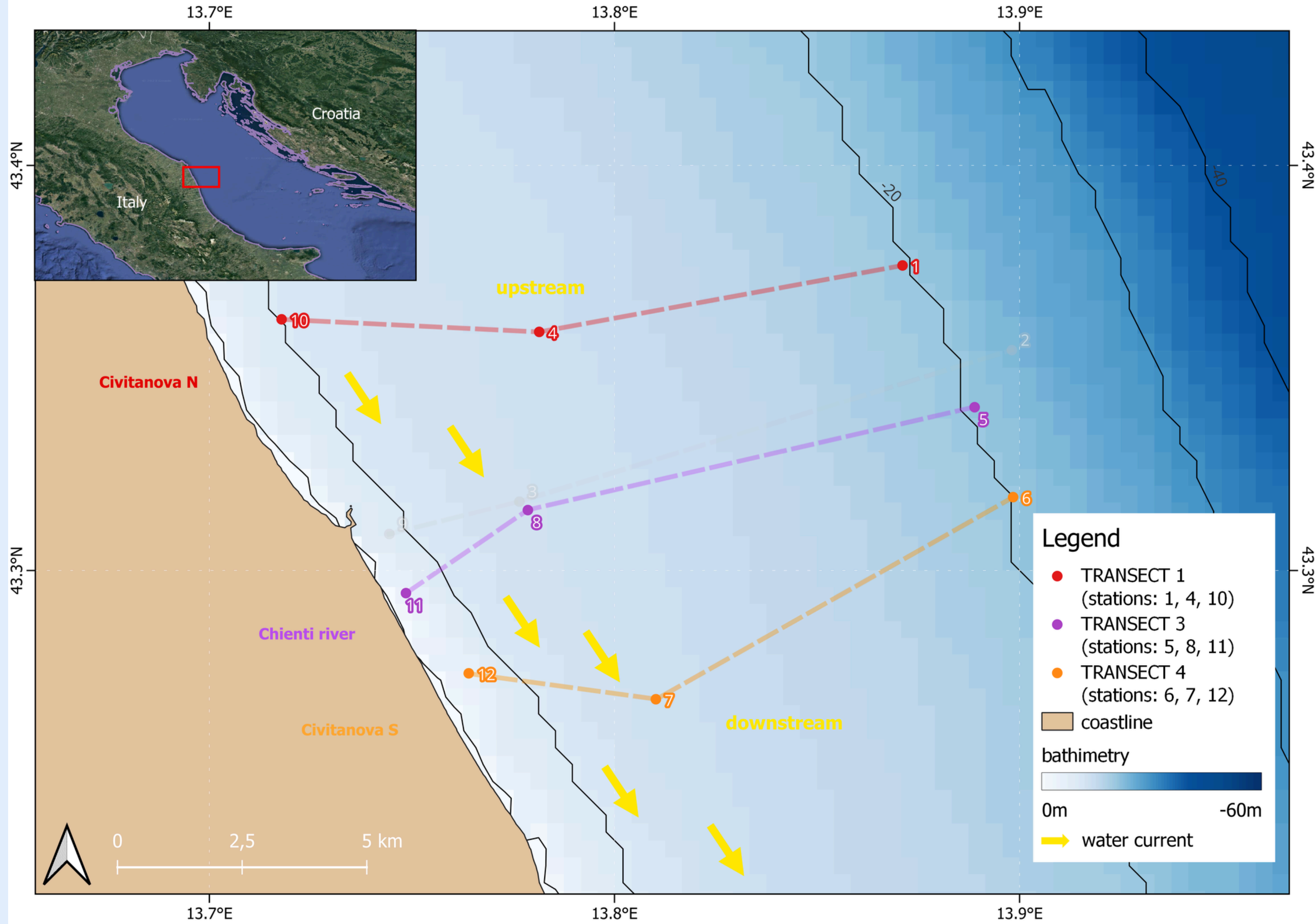
River runoff affects environmental quality and fauna, with stronger, asymmetrical impacts downstream due to coastal currents.



Study area



Study area



Sampling activities on board

Research Vessel:
ANNA GUIDOTTI

- **IMO number:** 9096686
- **Type:** Offshore Supply Ship
- **Length (LOA):** 30.99 meters
- **Width (beam):** 6 meters



20th February 2025

Methodologies



**Physical-Chemical
Parameters**



Biological Indicators
Water Column



Biological Indicators
Benthic Fauna

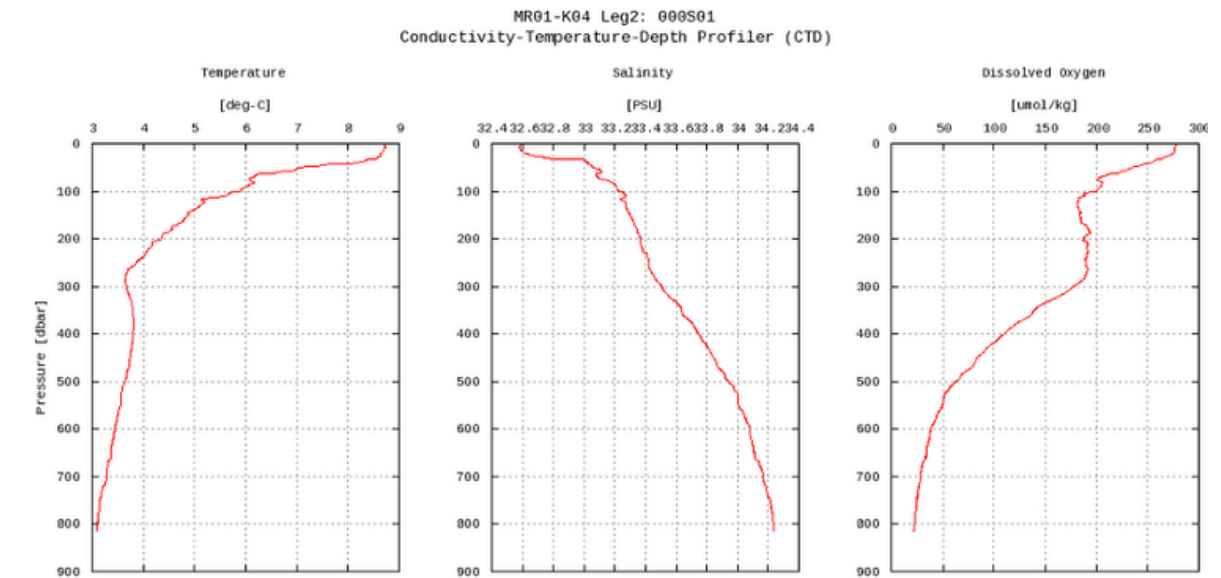
Methodologies



Physical-Chemical Parameters

1. CTD Profiling:

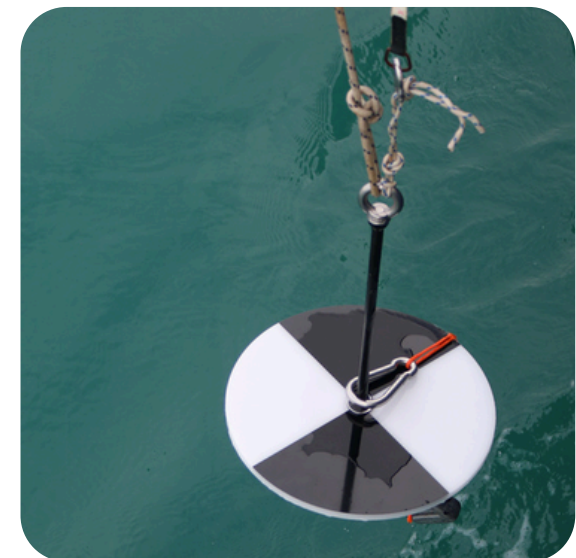
Measurement of Salinity, Temperature and Depth along the water column.



Biological Indicators *Water Column*

2. Turbidity:

Estimation of water transparency with Secchi Disk.



Biological Indicators *Benthic Fauna*

3. TRIX Index:

Trophic state index combining nutrients and Chl-a data.

Methodologies



**Physical-Chemical
Parameters**



Biological Indicators
Water Column



Biological Indicators
Benthic Fauna

1. Phytopigments (Chlorophyll-a):

Indicator of phytoplankton biomass.

2. Zooplankton Analysis:

Assessment of abundance, Shannon Index, and Pielou's Evenness.



Methodologies



**Physical-Chemical
Parameters**



Biological Indicators
Water Column



Biological Indicators
Benthic Fauna

1. Meiofauna:

Quantification of small benthic organisms (e.g., nematodes, copepods).

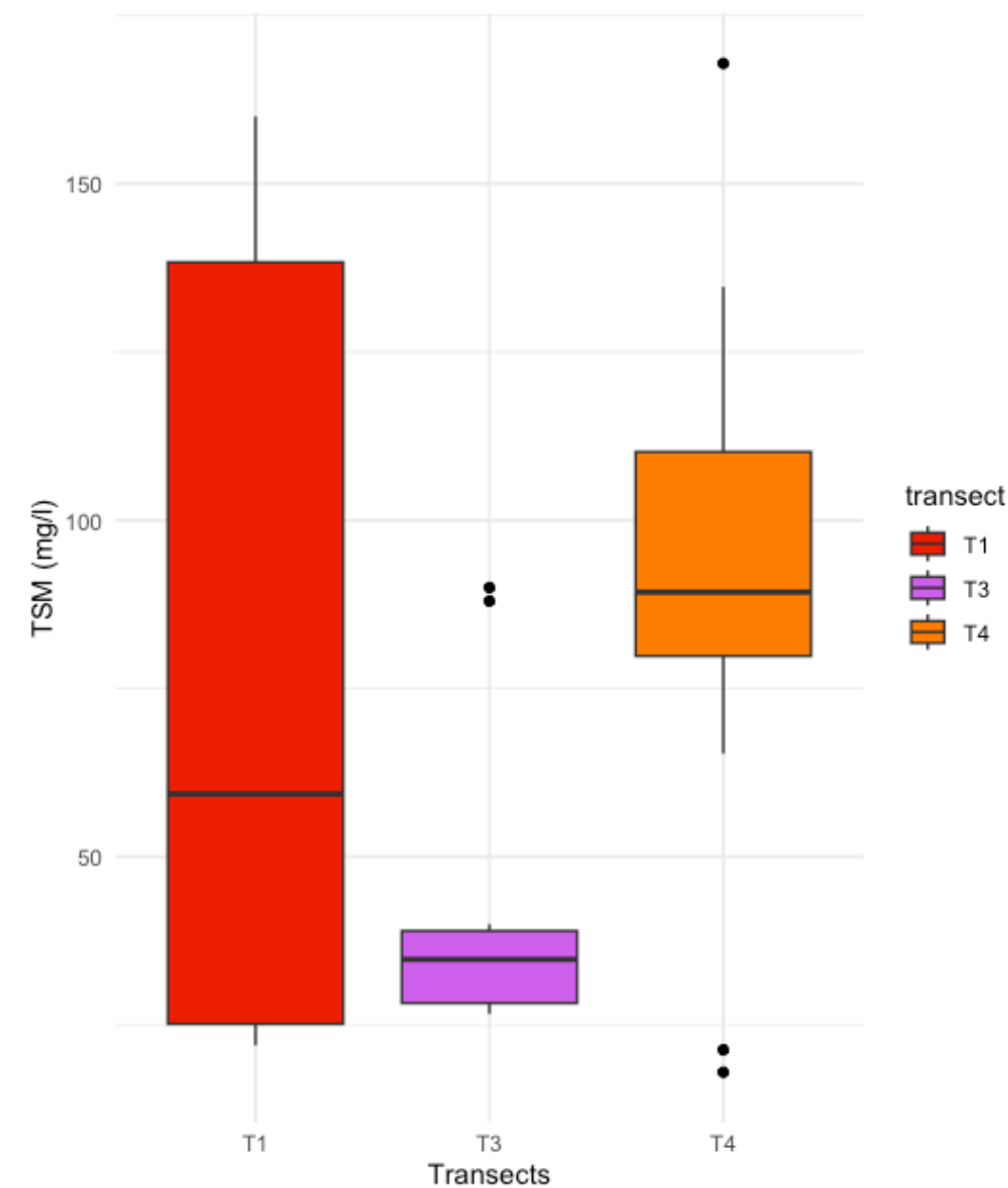
2. Macrofauna:

Community analysis based on abundance, Shannon Index, and Pielou Index.



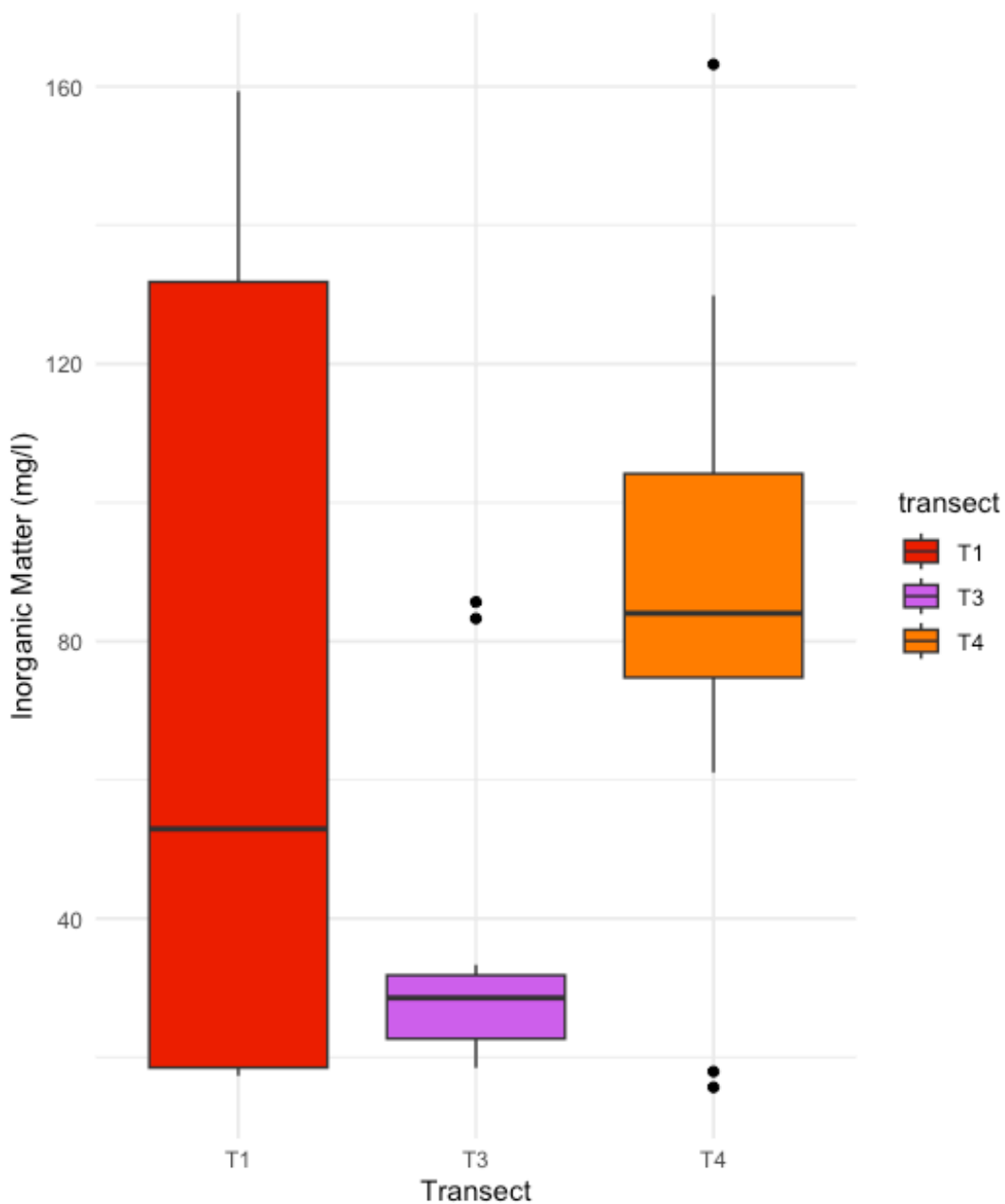
Water column

Total Suspended Matter



Tuckey post hoc
T4-T1 p-value 0.80
T4-T3 p-value 0.03 *

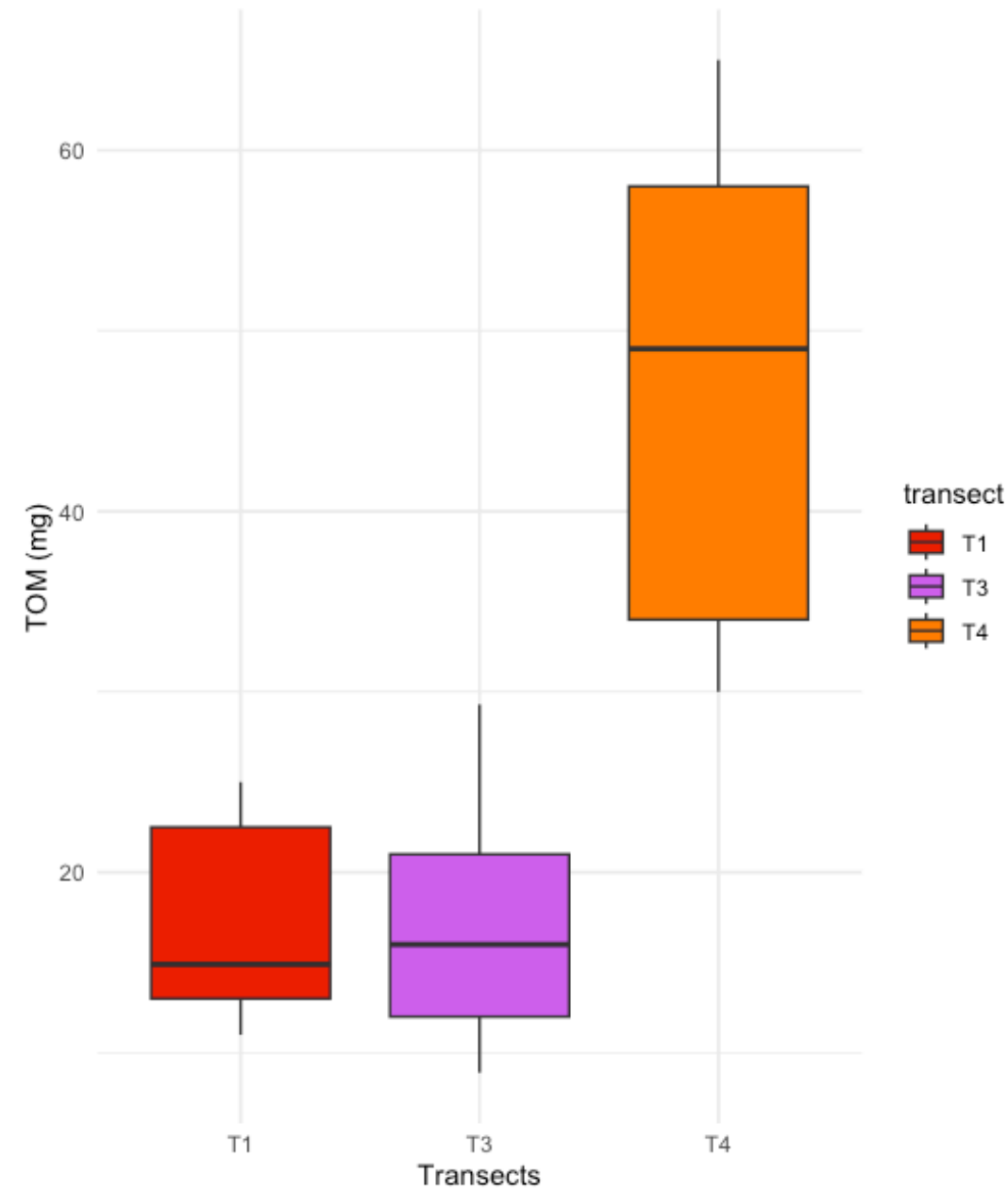
Inorganic Matter



Tuckey post hoc
T4-T1 p-value 0.78
T4-T3 p-value 0.03 *

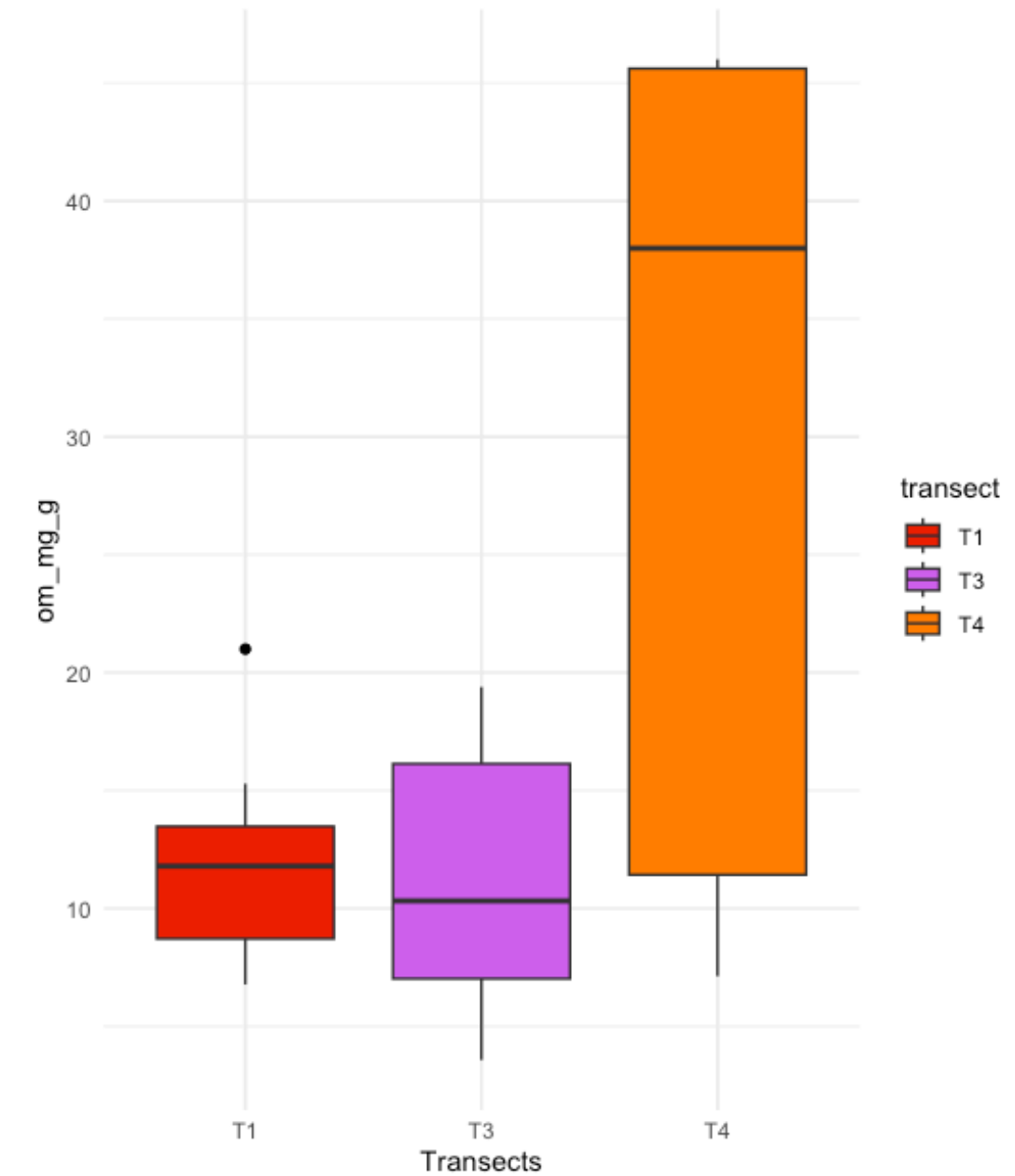
Sediment

Total Organic Matter



Tuckey post hoc
T4-T1 p-value 0,0000005 ***
T4-T3 p-value 0,0000004 ***

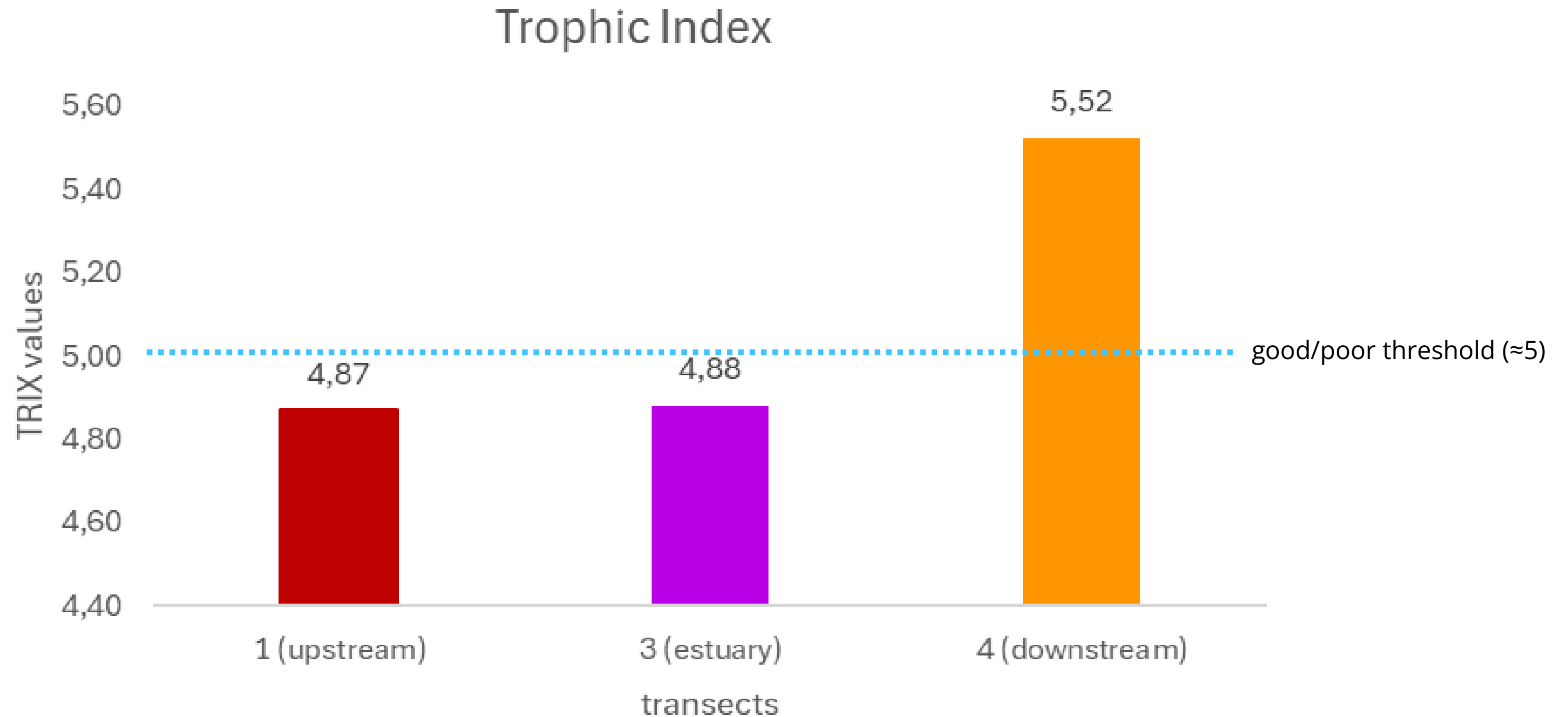
Organic Matter



Tuckey post hoc
T4-T1 p-value 0,0053 **
T4-T3 p-value 0,0041 **

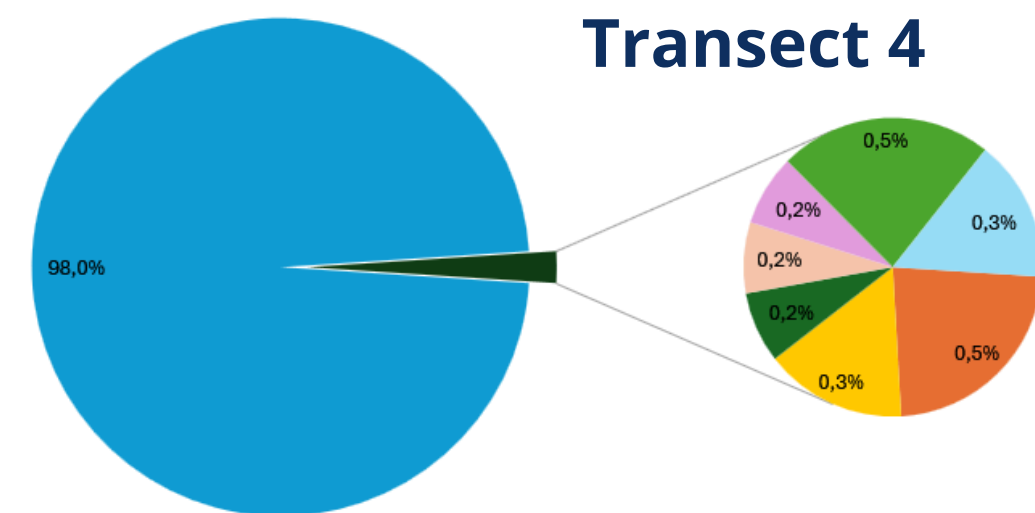
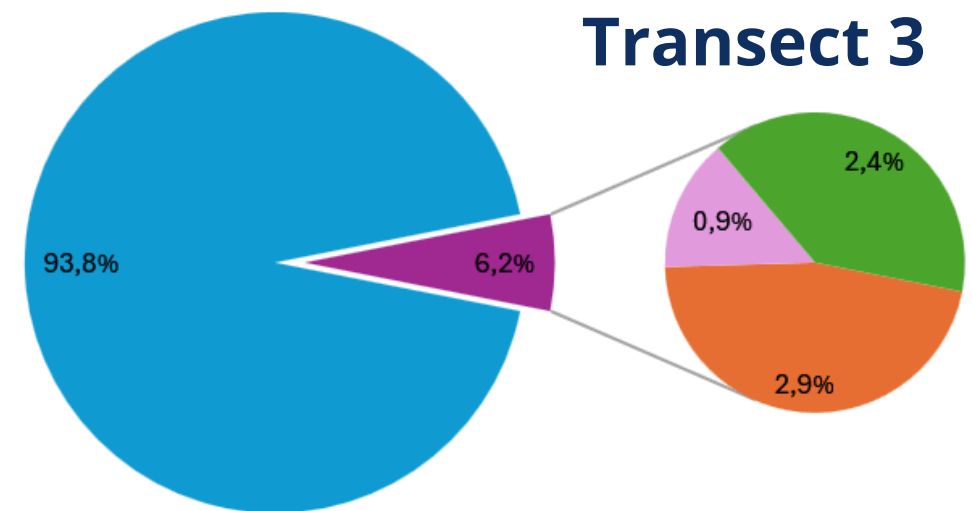
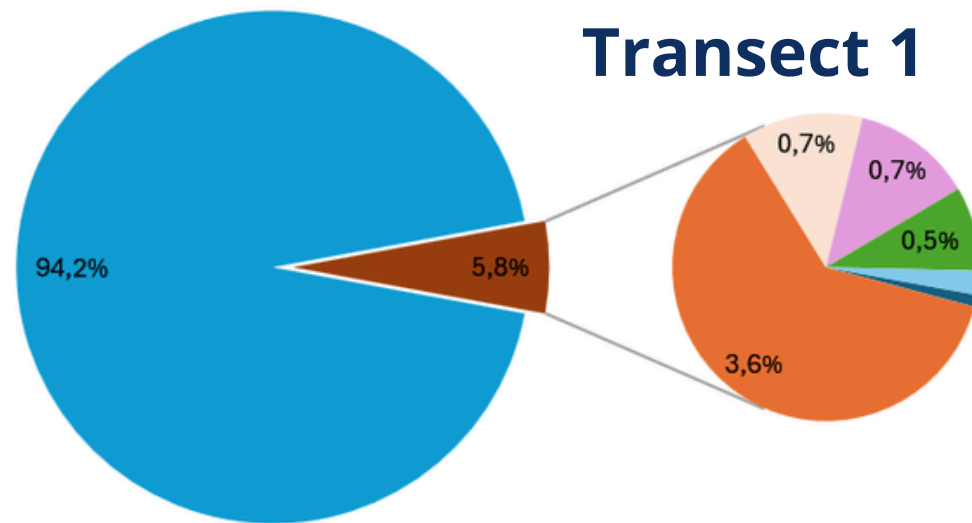
Trophic Index - TRIX

$$\text{TRIX} = [\text{Log10} (\text{Chl-a} \times \text{aDO} \times \text{N} \times \text{P}) + 1.5] / 1.2$$



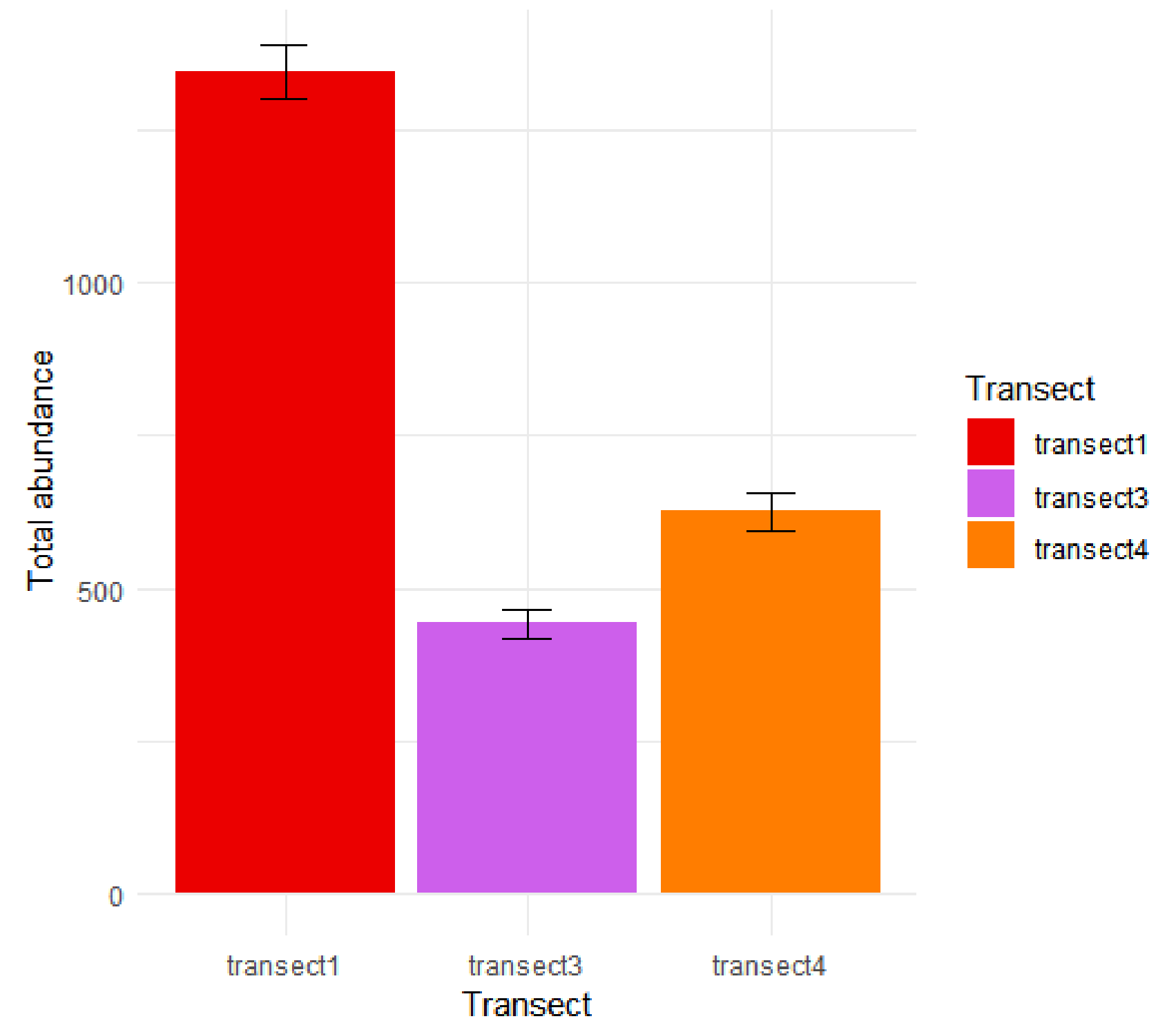
Meiofauna

Taxa distribution



- Copepoda
- Gastrotricha
- Isopoda
- Kinorinch
- Nematods
- Ostracoda
- Polychaeta
- Turbellaria
- Acarine

Total abundance of meiofauna



Meiofauna Taxa Richness

Transect 1 = 7 taxa

Transect 3 = 4 taxa

Transect 4 = 8 taxa

Non-polluted systems: 10 - 18 taxa

Polluted systems: 4 - 8 taxa

Classification proposal

<4 taxa $\Rightarrow$ bad conditions

4-8 taxa $\Rightarrow$ altered condition

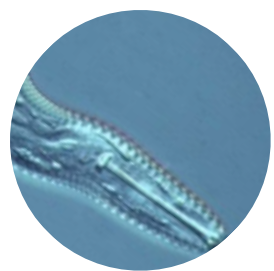
8-12 taxa $\Rightarrow$ sufficient conditions

12-16 taxa $\Rightarrow$ good conditions

> 16 taxa $\Rightarrow$ very good conditions

We can see an “**altered condition**” in all the analyzed transects but we don’t see an increased and asymmetrical impact due to coastal currents.

We have to take in mind that often Taxa Richness is not sufficient as pollution/impact indicator.



Nematodes/Copepodes Index



It is based on the greater tolerance of Nematodes to high organic loads and consequent reduced oxygen content with Copepodes which are much more sensitive to organic loads:

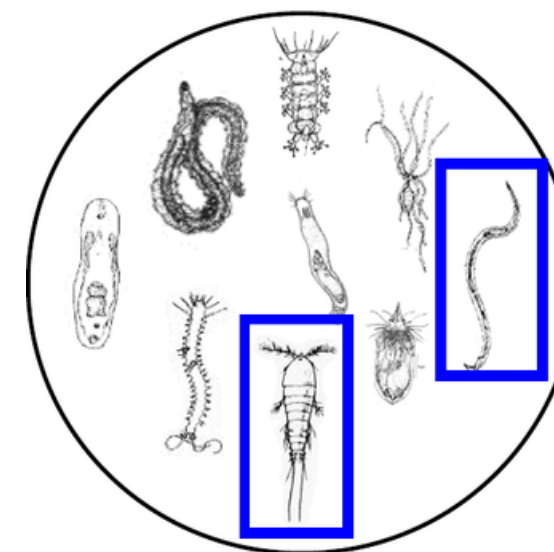
Transect 1 = N/C = 26.2

Transect 3 = N/C = 32.3

 **Transect 4 = N/C = 196.0**

Unimpacted environmental conditions

Nematodes / Copepods ≤ 10



Environments characterized by organic enrichment:

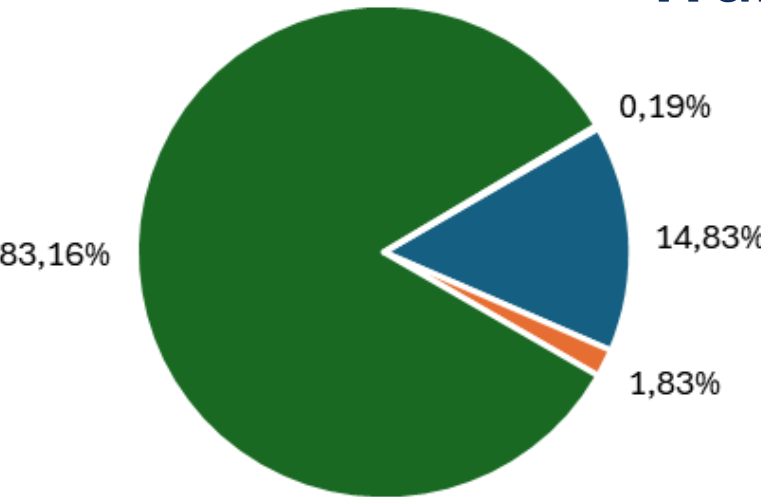
$20 \leq \text{Nematodes} / \text{Copepods} \leq 100$

In all the transects we can observe a **N/C typical of Environments characterized by organic enrichment**, especially we can observe a C/N ratio much higher in the Tr.4 than in Tr.1 and Tr.3, which could be a hint of greater current-driven organic load in that area.

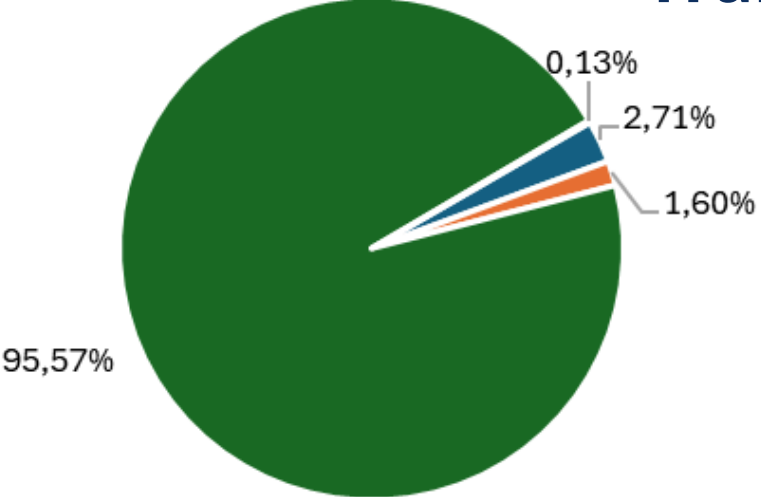
Macrofauna

Taxa distribution

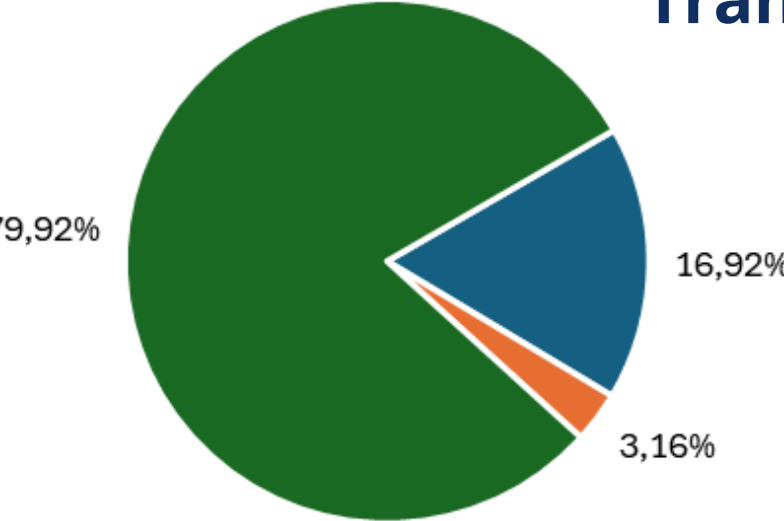
Transect 1



Transect 3

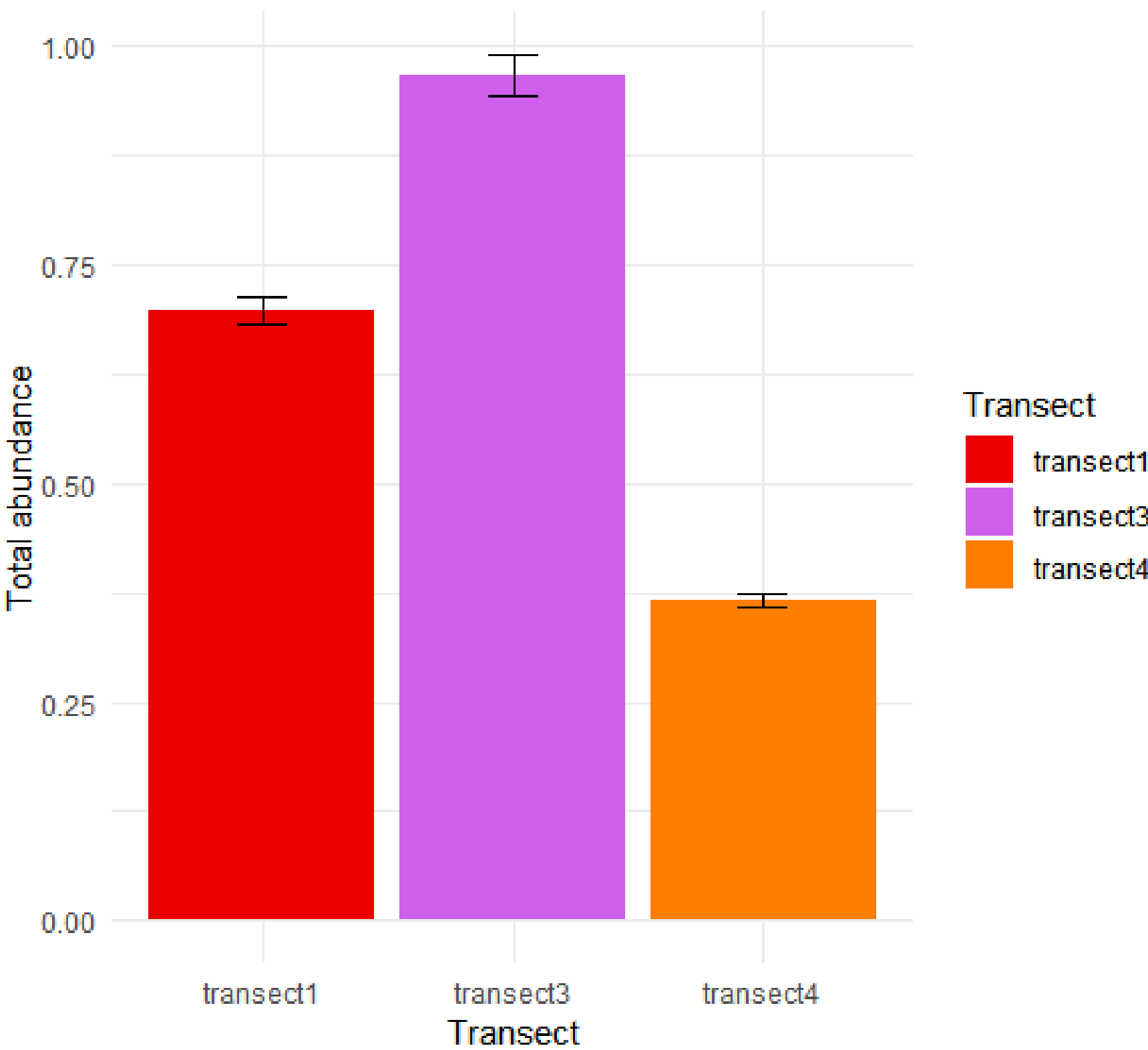


Transect 4



- Mollusks
- Crustacea
- Anellidae
- Cnidaria
- Echinoderms

Total abundance of macrofauna



Macrofauna Taxa Richness

Transect 1 = 5 taxa

Transect 3 = 5 taxa

Transect 4 = 4 taxa

Non-polluted systems: 10 - 18 taxa

Polluted systems: 4 - 8 taxa

Classification proposal

<4 taxa $\Rightarrow$ bad conditions

4-8 taxa $\Rightarrow$ altered condition

8-12 taxa $\Rightarrow$ sufficient conditions

12-16 taxa $\Rightarrow$ good conditions

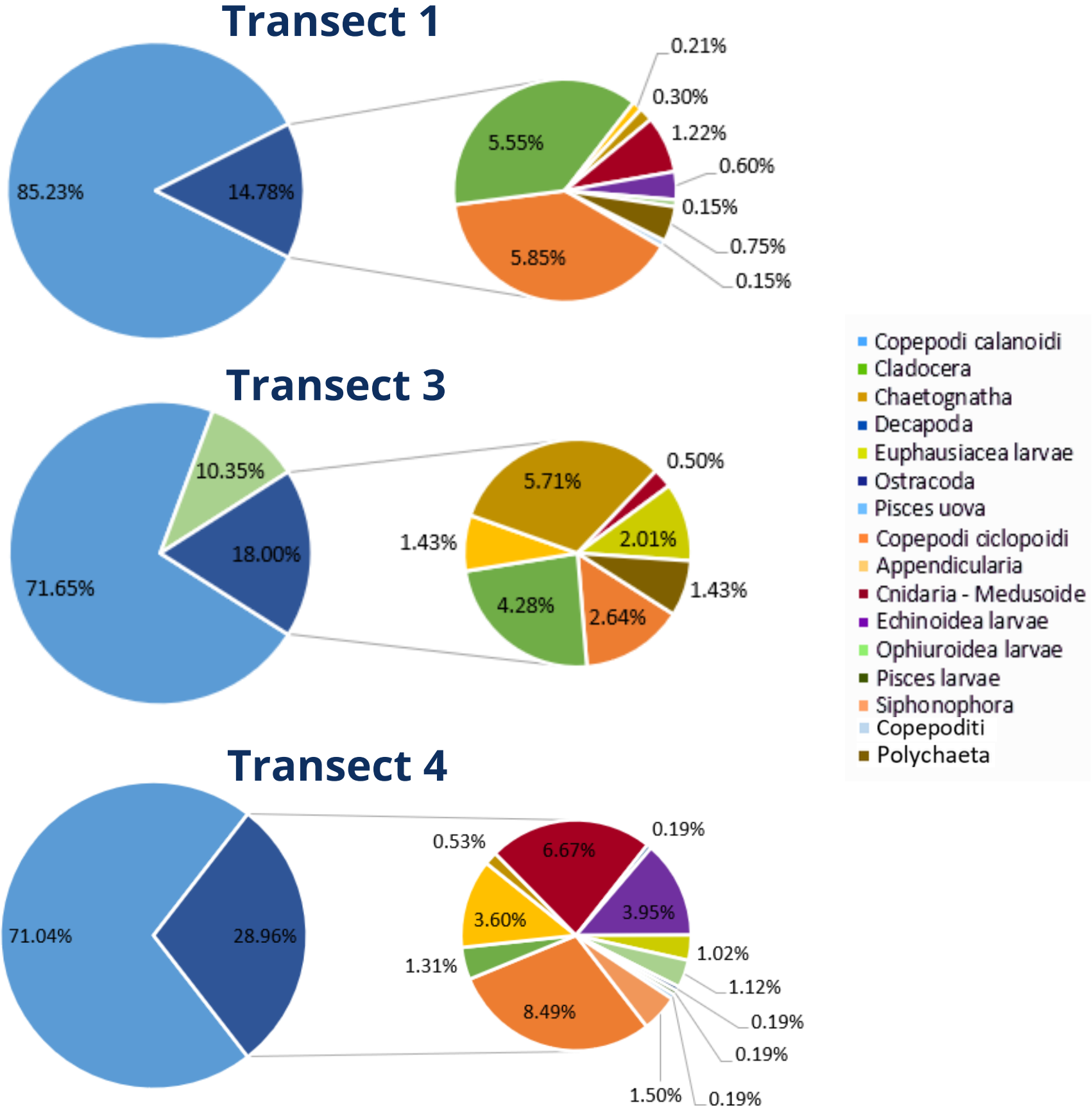
> 16 taxa $\Rightarrow$ very good conditions

We can see an “**altered condition**” in all the analyzed transects, with a very slight reduction of taxa in transect 4.

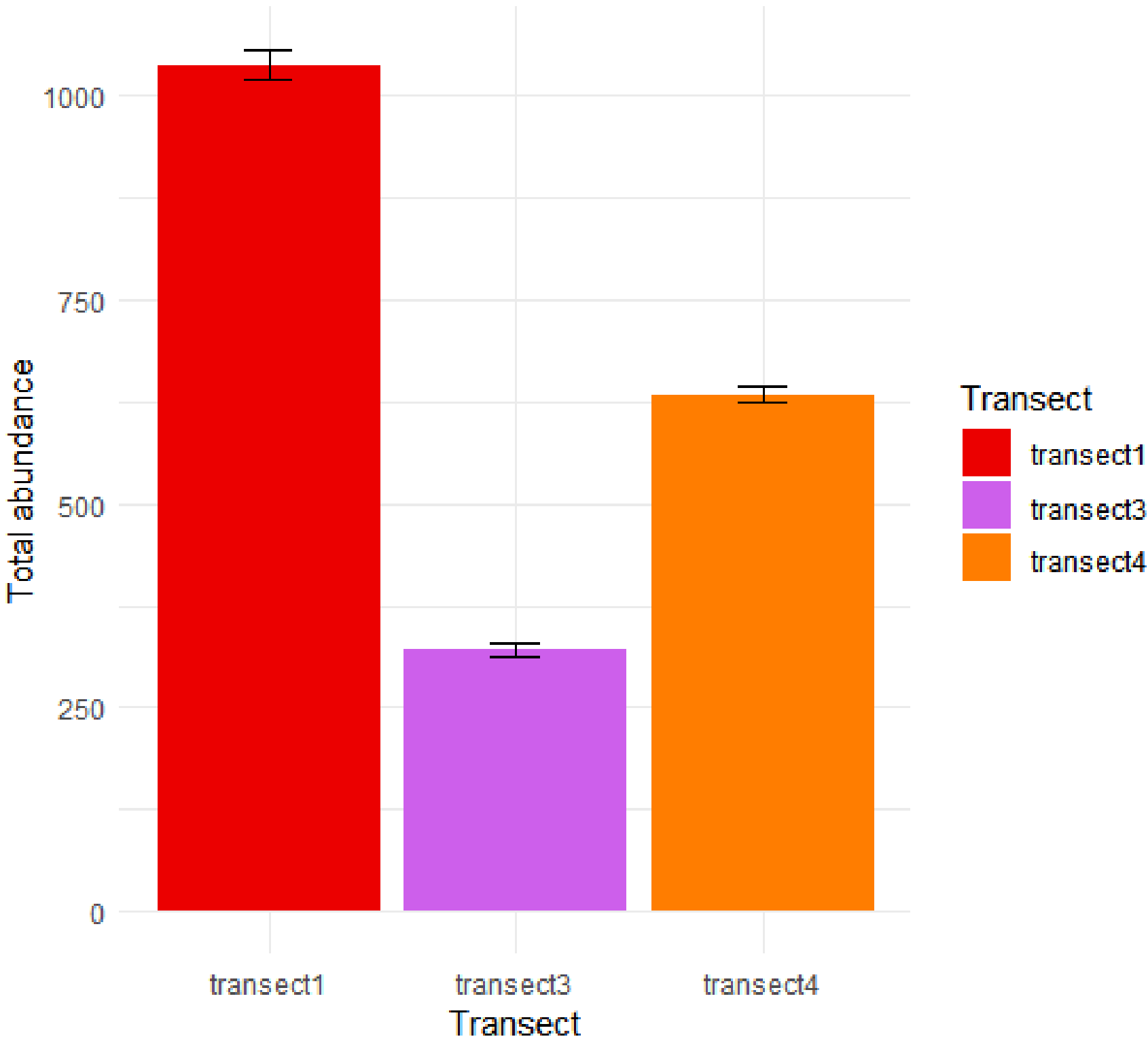
We have to take in mind that often Taxa Richness is not sufficient as pollution/impact indicator.

Zooplankton

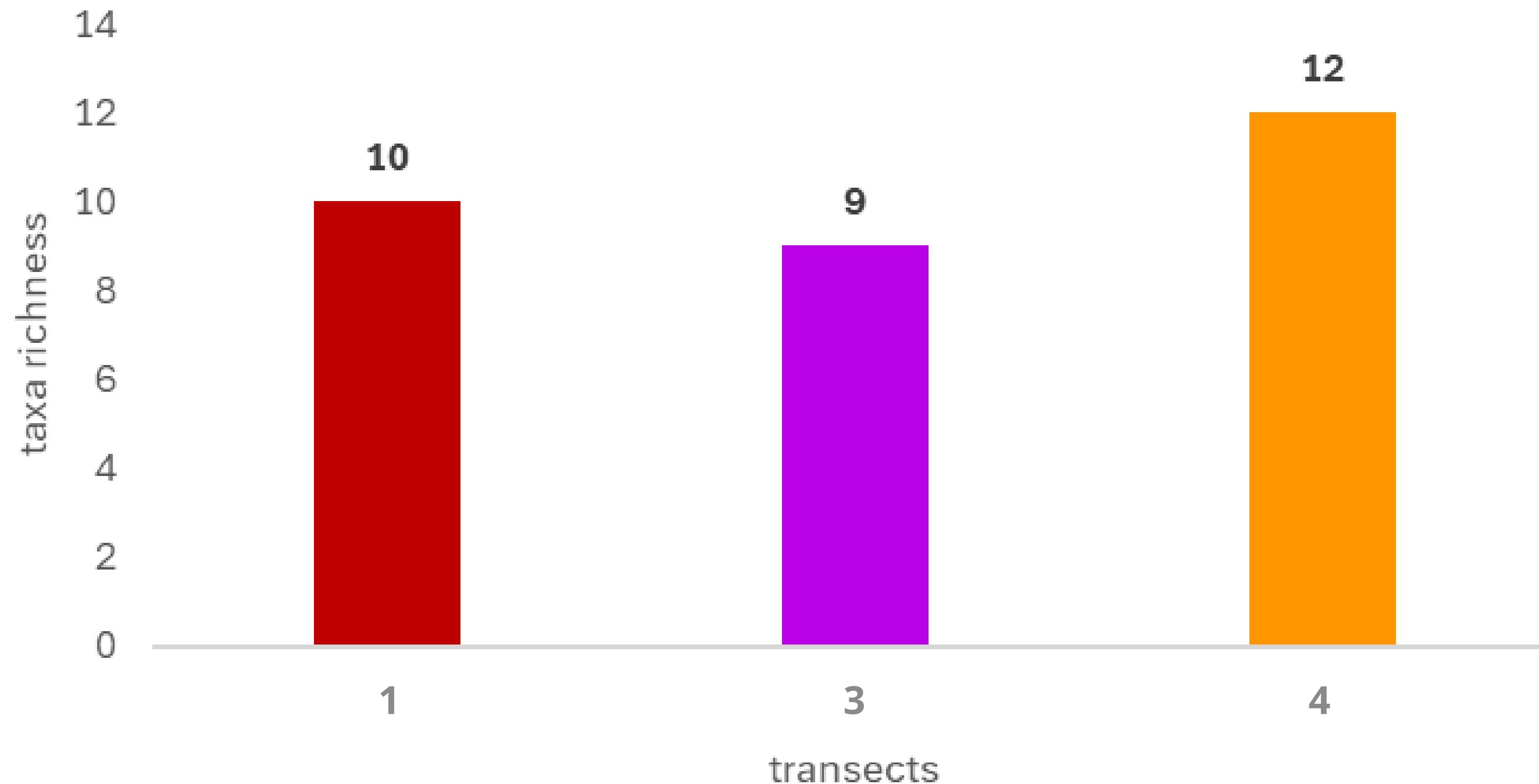
Taxa distribution



Total abundance of zooplankton

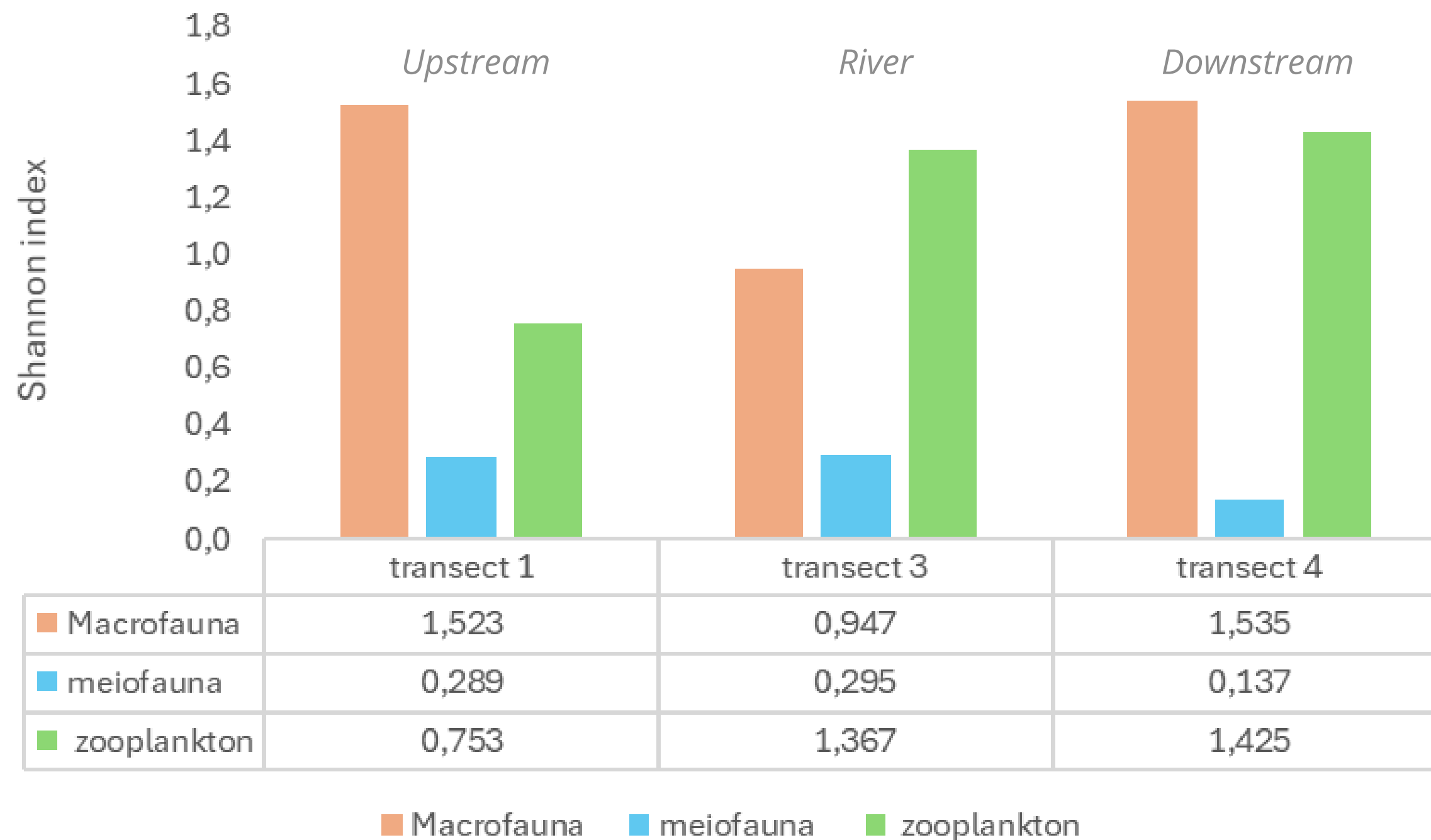


Zooplankton Taxa Richness



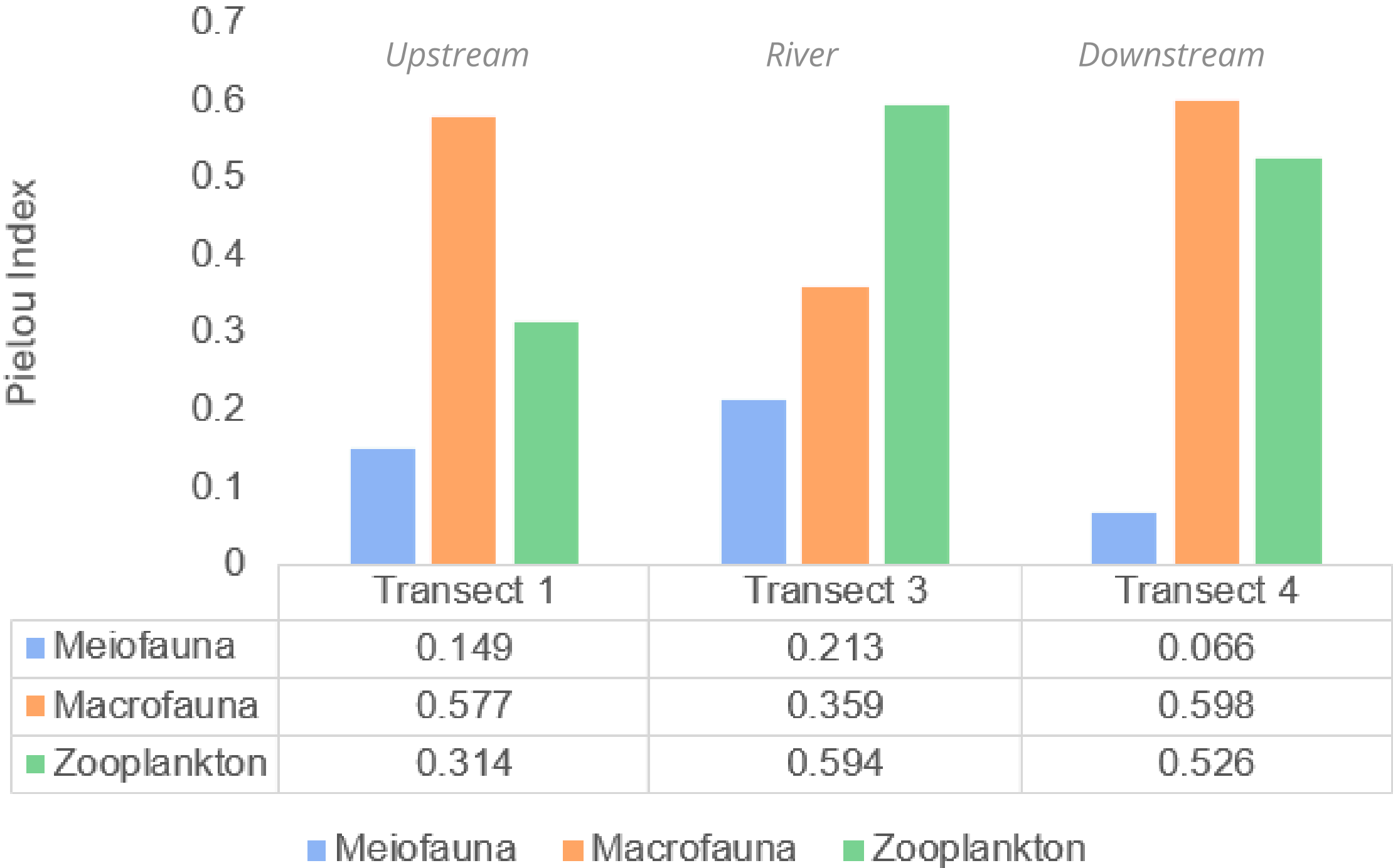
Zooplankton is very dynamic, sensitive to temporal, spatial, biotic and abiotic factors, so there are no universal, standardised reference thresholds for taxa richness as there are for benthic macrofauna and meiofauna.

Shannon-Wiener Index



- **MACROFAUNA: Tr. 3**
 → lowest macrofaunal diversity.
 → possible stress/disturbance.
- **MEIOFAUNA**
 → low overall diversity,
 nematode dominance.
- **ZOOPLANKTON: Tr. 3 - 4**
 → higher zooplankton diversity
 → possible link with increased
 phytoplankton due to
 nutrient enrichment.

Pielou's Evenness Index



- **MEIOFAUNA**
 - low evenness across all transects but lowest at Tr. 4.
 - short-term impact index.
- **MACROFAUNA**
 - lowest evenness at Tr. 3 (estuary).
 - long-term impact index.
- **ZOOPLANKTON**
 - same reasoning as Shannon index analysis.

To sum up

ANALYSIS	WATER	SEDIMENT
TRIX		—
TSM		—
IM		—
TOM	—	
OM		

To sum up

ANALYSIS	MEIOFAUNA	MACROFAUNA	ZOOPLANKTON
Total abundance	✓	✓	✓
Taxa richness	✗	✗	—
Ne/Co	✓	—	—
Shannon-Wiener Index	✓	✓	✓
Pielou's Evenness Index	✓	✗	✓

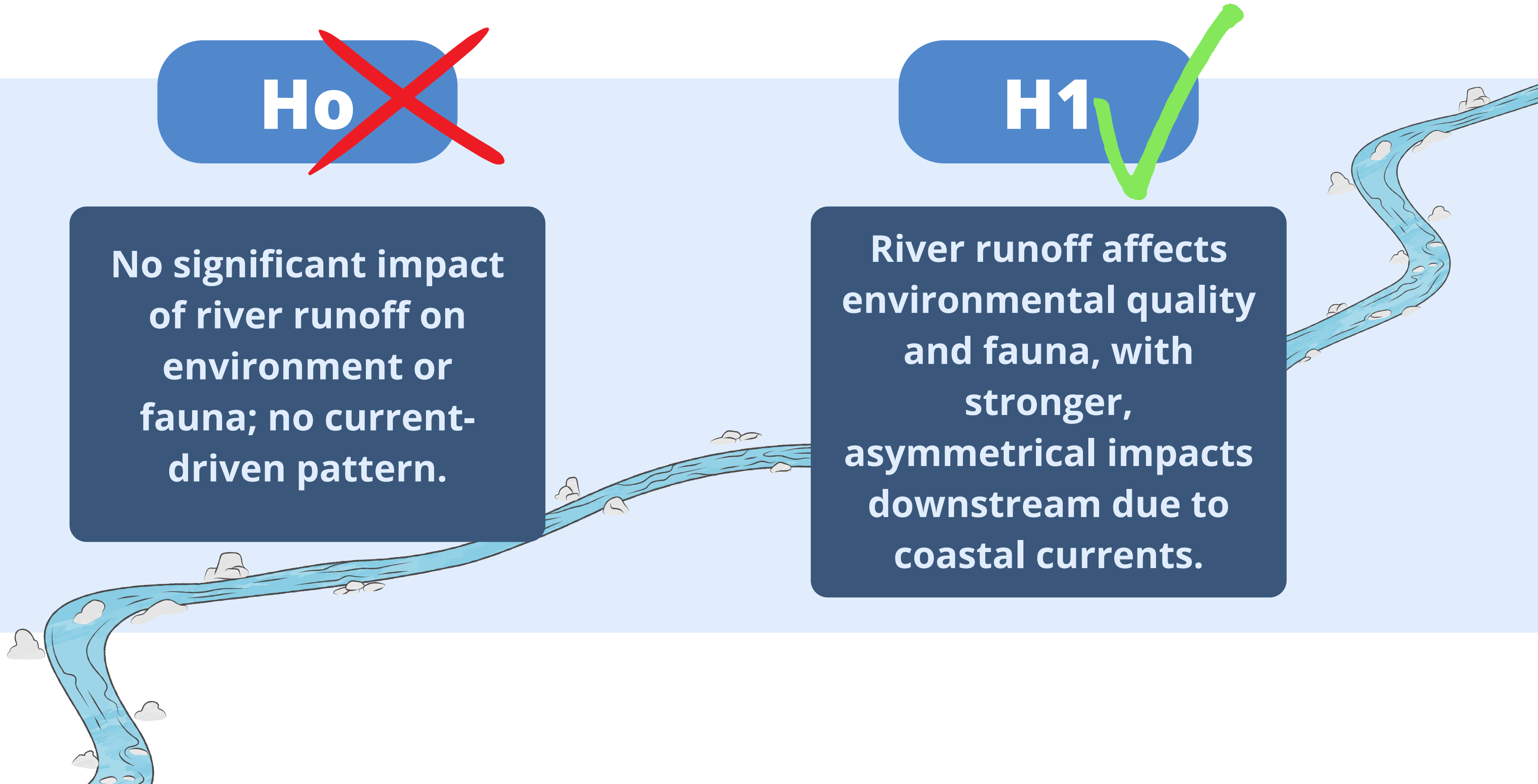
Hypotheses

Ho

No significant impact of river runoff on environment or fauna; no current-driven pattern.

H1

River runoff affects environmental quality and fauna, with stronger, asymmetrical impacts downstream due to coastal currents.



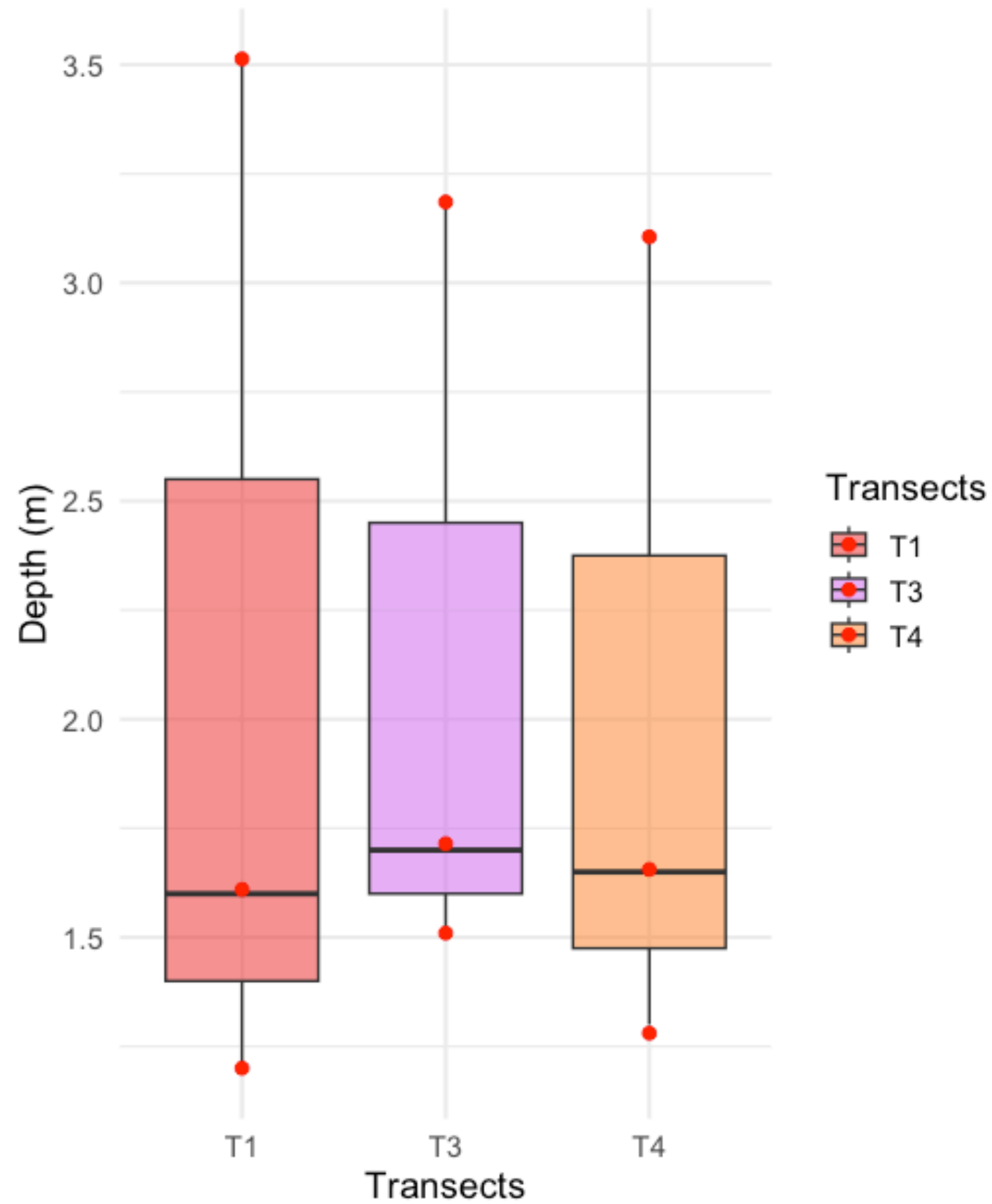


THANK YOU!

Bibliography and Sitography

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- SCURI, Stefania, et al. The European Water Framework Directive 2000/60/EC in the evaluation of the ecological status of watercourses. Case study: the river Chienti (central Apennines, Italy). *Acta hydrochimica et hydrobiologica*, 2006, 34.5: 498-505.
- <https://www.arpa.marche.it/indicatori-ambientali?id=1046>

Secchi Disk



P-value = 0.99 NO significant difference.